
ZIPPO PLATFORM ARCHITECTURE & SPECIFICATION

Spec-Driven Development (SDD) & Test-Driven Development (TDD) Guide

Generated: September 15, 2026 | Version: 1.0.0 | Status: APPROVED

* All external HTTP requests arrive at port 80 of the shared ALB and receive an immediate '3

* Path patterns evaluate against listener rules in order of priority:

* '/api/*' (Priority 10) forwards to 'zippo-be' target group.

* '/' (Priority 100) forwards to 'zippo-fe' target group.

* In-process sidecars (e.g., Redis container in 'zippo-be') communicate via 'localhost:<port'

* Building and caching base runner Docker images (e.g., node runner, terraform runner).

* Defining the container listening port ('PORT: 3000').

* Defining the health check endpoint ('HEALTH: "/api/v1/health"').

* Defining resource sizing ('CPU: 256', 'MEMORY: 1024', 'DESIRED_COUNT: 1').

* Declaring required sidecar containers (e.g. 'redis-alpine').

* Enabling AWS SES access ('SES: true') if the service sends transactional email.

* Specifying S3 bucket access ('S3_BUCKET: "zippo-files"') if the service handles file uploa

* Maintaining a secure, multi-stage 'Dockerfile'.

* Executing unit, integration, and lint tests in the CI workflow.

* Pushing built images to ECR tagged with '<commit-sha>-<timestamp>'.

* Updating the SSM release tag parameter ('/ci/<service-name>/image_tag').

* Executing 'terraform apply' on the app's 'deploy/<env>' directory.

-
- **Runner:** Ephemeral AWS CodeBuild container ('codebuild-ZIPPO-<APP>-<run_id>').

• **Actions:**

* Dependency installation with cache restoration ('~/npm').

* Static code analysis & linting ('npm run lint').

• ****Actions:****

* Resolve AWS Account ID dynamically via 'aws sts get-caller-identity'.

* Compute release tag: 'IMAGE_TAG="\${SHORT_SHA}-\${DATE_TAG}"' (e.g. '41d0162-15-09-26-12-00')

* Authenticate Docker daemon against Amazon ECR:

""bash

* Build container using root 'Dockerfile'.

-
- ****SSM Parameter Store Key:**** '/ci/<service-name>/image_tag'

• ****Command:****

“bash

```
aws ssm put-parameter --name "/ci/${SERVICE_NAME}/image_tag" --value "${IMAGE_TAG}" --overwri
```


-
- **Rationale:** Decouples image production from infrastructure updates. Terraform reads thi

- ****Path:**** 'deploy/<env>/'

-
- **Module Resolution:** Configures Git credentials to fetch the private module 'ZIPPO-INFR/

• ****Commands:****

“bash

-
- ECS registers a new revision of the task definition with the new image tag.

-
- Fargate starts new tasks, runs health checks against the target group, verifies healthy st

* Infrastructure state is isolated in S3 ('zippo-terraform-state-bucket', key 'zippo' or 'zi

""bash

1. Developer triggers 'workflow_dispatch' with:

* Service ('zippo-be', 'zippo-fe', etc.)

2. Workflow creates the parameter or secret in AWS with value "PLACEHOLDER".

1. ****Service Identifier:**** What is the lowercase kebab-case name of the service? (e.g., 'zippo-

2. **Repository Location:** GitHub organization and repo name (e.g., 'epddp/ZIPPO-BILLING').

3. ****Owning Team & Tech Lead:**** Who is responsible for maintaining this application?

5. ****Listening Port:**** What TCP port does the application listen on? (e.g., '3000', '8080', '80

10. ****ALB Routing Priority:**** What integer priority should this rule have? (Must be unique acro

11. ****Direct Port Requirement:**** Does this service require its own dedicated external HTTPS por

* Sidecar container name (e.g. 'redis', 'log-shipper').

13. ****Environment Variables:**** Provide a list of all non-sensitive configuration keys and their

* Example: 'LOG_LEVEL' -> '/zippo-billing/log_level' (String)

* Example: 'PAYMENT_GATEWAY_URL' -> '/zippo-billing/gateway_url' (String)

14. ****Encrypted Environment Variables:**** Provide a list of parameters requiring KMS decryption:

* Example: 'WEBHOOK_SIGNING_KEY' -> '/zippo-billing/webhook_signing_key' (SecureString: tru

15. **Secrets Manager References:** Provide the names of secrets stored in AWS Secrets Manager:

-
- 'main.tf': Instantiates 'module "aws_epam_ecs"' targeting the new environment backend stat

-
- 'terraform.tfvars': Declares subnet IDs, ALB security group, ACM domain, and CodeBuild pro

-
- 'variables.tf': Inherits standard variables.

Run 'terraform apply' to provision:

-
- The environment ECS cluster ('zippo-<env>-ecs' or 'cheap-ecs').

-
- The environment ALB with HTTPS listener and ACM certificate.

-
- The environment CodeBuild runner projects and webhooks.

For each active microservice ('ZIPPO-BE', 'ZIPPO-FE', etc.):

-
- Create 'deploy/<env>/config.yaml' populated with environment-specific URLs, database names

-
- Create 'deploy/<env>/main.tf', 'locals.tf', 'providers.tf', 'variables.tf', and 'versions.

-
- Populate secrets in AWS Secrets Manager and parameters in AWS SSM.

-
- Trigger initial deployment.

-
- ****Configuration Drift:**** Missing health checks, incorrect ALB priorities, or improperly fo

-
- **Security Regressions:** Missing permissions boundaries, insecure environment variables,

-
- **Context Hallucinations:** AI assistants guessing non-existent parameters or altering cri

-
- The schema definition is stored in 'zippo-specs/schemas/app-config.schema.json'.

-
- Validation is executed locally or in CI via:

“bash

-
- No Terraform code may be planned or applied if schema validation fails.

â satisfy the assertion. Test passes instantly. â

-
- **Mock Providers:** `mock_provider "aws"` mocks all API calls, preventing AWS credential d

-
- **Command Mode:** Running with 'command = apply' or 'command = plan' checks HCL logic, con

-
- **Instant Feedback:** Tests execute in under 3 seconds.

